

Interrogating basal ganglia circuit function in Parkinson's disease and dystonia

Reviewed Preprint

v2 • July 30, 2024

Revised by authors

Reviewed Preprint

v1 • November 20, 2023

Srdjan Sumarac, Kiah A Spencer, Leon A Steiner, Conor Fearon, Emily Haniff, Andrea A Kühn, Mojgan Hodaie, Suneil K Kalia, Andres M Lozano, Alfonso Fasano, William D Hutchison, Luka Milosevic 

Krembil Brain Institute, University Health Network, Canada • Institute of Biomedical Engineering, University of Toronto, Canada • Department of Neurology and Experimental Neurology, Charité-Universitätsmedizin Berlin, Germany • Berlin Institute of Health (BIH), Germany • Edmond J. Safra Program in Parkinson's Disease, Morton and Gloria Shulman Movement Disorders Clinic, Toronto Western Hospital, Canada • Department of Neurology, University of Toronto, Canada • Institute of Medical Sciences, University of Toronto, Canada • Center for Advancing Neurotechnological Innovation to Application (CRANIA), Toronto, Canada • Department of Surgery, University of Toronto, Canada • KITE, University Health Network, Toronto, Canada • Department of Physiology, University of Toronto, Canada

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Background

The dichotomy between the hypo-versus hyperkinetic nature of Parkinson's disease (PD) and dystonia, respectively, is thought to be reflected in the underlying basal ganglia pathophysiology.

Objective

Investigate differences in globus pallidus internus (GPI) neuronal activity, and short- and long-term plasticity of direct pathway projections.

Results

GPI neurons were slower, burstier, and less regular in dystonia. In PD, symptom severity positively correlated with the power of low-beta frequency spiketrain oscillations. In dystonia, symptom severity negatively correlated with firing rate, and positively correlated with neuronal variability and the power of theta frequency spiketrain oscillations. Dystonia was moreover associated with less long-term plasticity and slower synaptic depression.

Conclusion

We substantiated claims of hyper-versus hypofunctional GPi output in PD versus dystonia, and provided cellular-level validation of the pathological nature of theta and low-beta oscillations in respective disorders. Such circuit changes may be underlain by disease-related differences in plasticity of striato-pallidal synapses.

eLife assessment

This manuscript is a **valuable** study of the responses of GPi neurons to deep brain stimulation (DBS) in human Parkinson disease and dystonia patients and it finds **convincing** evidence for altered short-term and long-term plasticity in response to DBS between the two patient populations. This dataset is of interest to both basic and clinical researchers working in the field of DBS and movement disorders.

<https://doi.org/10.7554/eLife.90454.2.sa3>

Introduction

Parkinson's disease (PD) is a complex neurodegenerative disorder characterized by non-motor manifestations and motor symptoms including tremor, bradykinesia, rigidity, postural instability, and freezing of gait [1]. Dystonia is a movement disorder characterized by involuntary muscle contractions, repetitive movements, and abnormal fixed postures [2]. While PD is characterized as a collection of hypokinetic syndromes (apart from tremor and levodopa-induced dyskinesias), dystonia is considered a hyperkinetic movement disorder.

The dichotomy between the hypo-versus hyperkinetic nature of PD and dystonia, respectively, is thought to be reflected in the underlying basal ganglia pathophysiology. In particular, basal ganglia circuitry that involves “direct” (D1-mediated striatal projections to the internal globus pallidus; GPi) and “indirect” (D2-mediated striatal projections via the external globus pallidus and subthalamic nucleus to GPi) pathways has been implicated in these disorders [3]. In PD, activity in the indirect pathway is upregulated while the direct pathway is downregulated, both of which converge towards increased GPi-mediated inhibition of thalamocortical motor networks [4]. Conversely, dystonia has been hypothesized to occur as a result of hypofunctional indirect pathway activity [5,6] and/or hyperfunctional direct pathway activity [7], producing decreased GPi-mediated inhibition of thalamocortical motor networks. These circuit differences would thereby result in increased GPi firing rates in PD and decreased rates in dystonia; which has been substantiated by some [8,9] but not all [10] human single-neuron studies. In addition to potential differences in single-neuron firing properties, each of the disorders has been associated with aberrant synchronization at the neural aggregate level; namely, increased beta (13-30Hz) and theta (4-12Hz) local field potential (LFP) oscillations in PD [11] and dystonia [12], respectively.

Despite opposing neurocircuit pathologies, deep brain stimulation (DBS) of the GPi is a widely used treatment for the motor symptoms of both PD and dystonia. However, the timescales of symptomatic improvement after GPi-DBS and the reappearance of symptoms upon stimulation cessation are markedly longer in dystonia (minutes-hours-days) compared to PD (seconds-minutes) [13]. These differences in timescales may be the result of disease-specific dynamics of

various forms of synaptic plasticity [14]. Intriguingly, high-frequency GPi stimulation has been shown to induce synaptic depression of inhibitory striato-pallidal (GPi) projections during stimulation (short-term plasticity effects), followed by an enduring potentiation of these projections after stimulation cessation (long-term plasticity-like effects) in both patients with PD [15] and dystonia [16]. However, the dynamics of these plastic effects have not been directly compared between disorders.

In this work, we compared GPi single-neuron activities (rate and oscillation -based features) between PD and dystonia, and investigated the relationships between electrophysiological features and disease severity. Moreover, we investigated differences in the dynamics of short- and long-term plasticity of the direct pathway projections.

Methods

Patients and clinical scores

For the single-neuron feature analysis, data were collected from 19 patients with non-genetic dystonia (135 included segments; see Supplementary Materials for information on subtypes) and 44 patients with PD (222 segments) during awake microelectrode-guided GPi-DBS surgeries [17]; after overnight withdrawal of antiparkinsonian medications in patients with PD. Corresponding preoperative clinical scores for PD using the Unified Parkinson's Disease Rating Scale Part III (UPDRSIII) and dystonia using either the Burke-Fahn-Marsden Dystonia Rating Scale (BFMDRS) or the Toronto Western Spasmodic Torticollis Rating Scale (TWSTRS) were also amalgamated for each patient. Given that UPDRSIII includes both hypokinetic and hyperkinetic symptoms of PD, we further sought to disaggregate the score by only considering items 23-26 in UPDRSIII, which assess hypokinetic symptoms of PD. To enable comparison across different dystonia scales, clinical scores were normalized (0-100%) with respect to the maximum severity per scale (108 for total UPDRSIII, 32 for hypokinetic only UPDRSIII, 85 for TWSTRS, and 120 for BFMDRS). For plasticity analyses, data were collected from a subset of 8 patients with dystonia (12 recording sites) and 10 patients with PD (13 recording sites). A data summary is available in Supplementary Table 1. Each patient provided informed consent and experiments were approved by the University Health Network Research Ethics Board and adhered to the guidelines set by the tri-council Policy on Ethical Conduct for Research Involving Humans.

Intraoperative data acquisition

Two independently driven microelectrodes (600µm spacing; 0.2–0.4MΩ impedances; ≥10kHz; sampling rate) were used for recordings; amplified using two Guideline System GS3000 amplifiers (Axon Instruments, Union City, USA) and digitized using a CED1401 data acquisition system (Cambridge Electronic Design, Cambridge, UK). For single-neuron data, units were sampled across the spatial extent of GPi as part of the standard of care neurophysiological mapping procedure; as previously described [17]. For plasticity data, microstimulation was delivered from one microelectrode using biphasic (cathodal followed by anodal) pulses (100µA, 150µs) from an isolated constant current stimulator (Neuro-Amp1A, Axon Instruments), while recording field evoked potentials (fEPs) using an adjacent microelectrode at the same depth; as done previously [15, 16, 18]. Initially, ten low-frequency stimulation (LFS) pulses were delivered at 1Hz to obtain baseline measurements of fEP amplitudes (i.e., functional readouts striato-pallidal inhibitory efficacy). Then, a standard high-frequency stimulation (HFS) tetanizing protocol was delivered (four 2s blocks of 100Hz, each separated by 8s) [16], after which another set of LFS pulses was delivered to obtain measurements of post-HFS fEP amplitudes (plasticity protocol depicted in Fig. 2A). Long-term plasticity analysis involved quantifying fEP changes before versus after HFS. Short-term plasticity analysis involved quantification of successive fEP amplitudes during HFS.

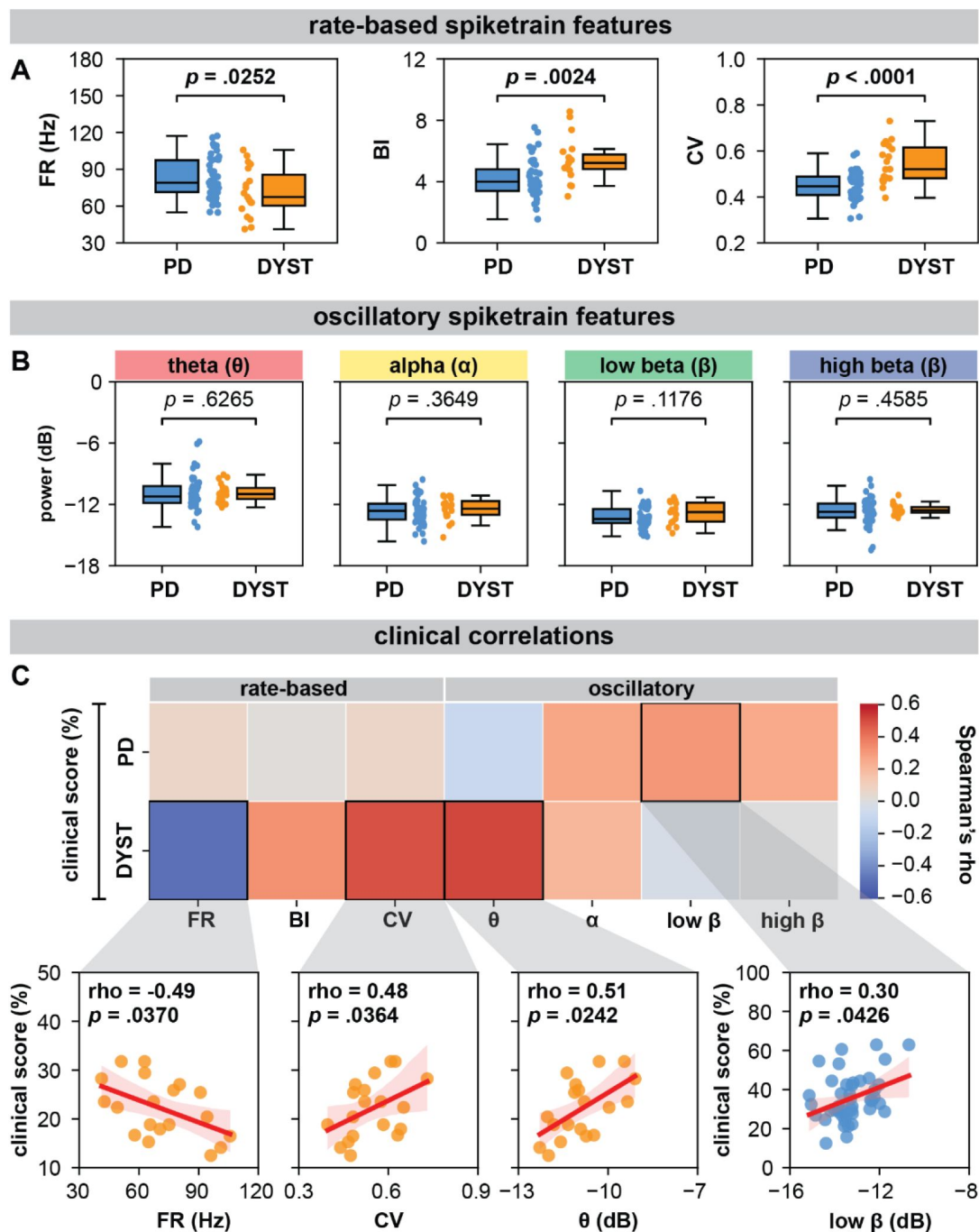


Fig 1.

Gpi spiketrain feature analyses and clinical correlates of PD and dystonia.

(A) With respect to (A) rate-based spiketrain features, firing rate was greater in PD while burst index (BI) and coefficient of variation (CV) were greater in dystonia; whereas no differences were found for (B) oscillatory spiketrain features for theta, alpha, low beta, high beta frequencies. MWU statistical results depicted are not corrected for multiple comparisons; after correction using the Bonferroni method, only CV and BI results remain significant (please see Supplementary Table 3). (C) In PD, the power of low beta spiketrain oscillations positively correlated (Spearman correlation) with symptom severity; in dystonia, neuronal firing rate negatively correlated with symptom severity, whereas CV and the power of theta spiketrain oscillations positively correlated with symptom severity. Depicted scatterplots are results that were significant before correction for multiple comparisons; however, none of the results persist after Benjamini-Hochberg correction for false discovery rate (please see Supplementary Table 4).

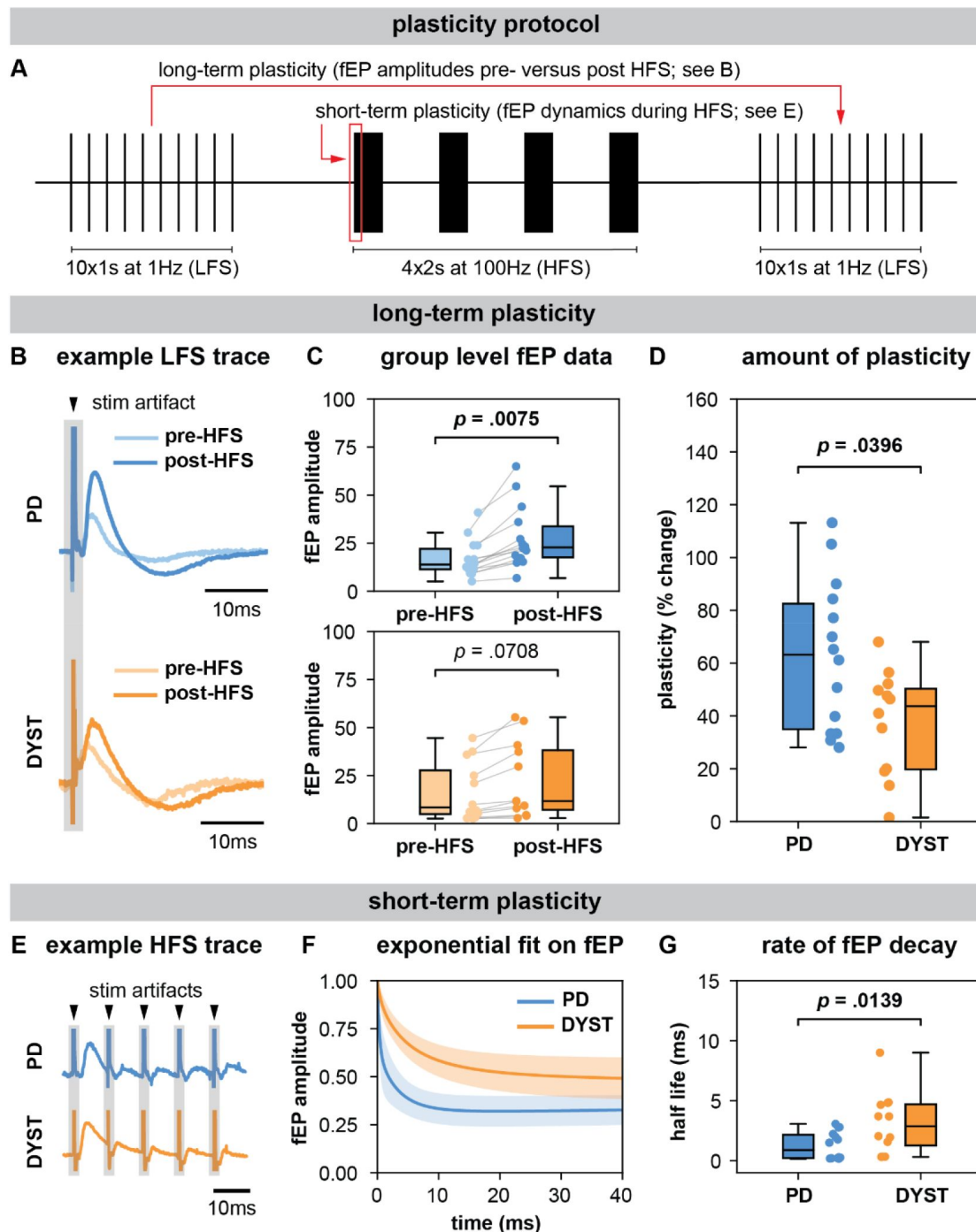


Fig 2.

Long-term and short-term effects of HFS on striato-pallidal plasticity in PD and dystonia.

(A) Schematic of the plasticity protocol to assess long-term plasticity via fEP amplitude comparisons pre-versus post-HFS and short-term plasticity via fEP dynamics during HFS. (B) Highlights example fEP traces for measuring long-term plasticity pre-versus post-HFS, with (C) displaying group-level fEP amplitudes pre-versus post-HFS across diseases. (D) Illustrates the amount of plasticity (i.e., percentage change in fEP amplitudes pre-versus post-HFS) in both PD and dystonia, with PD showing higher levels of plasticity. (E) Provides an example of fEP traces during HFS for assessing short-term plasticity, with (F) depicting group-level decay rates of fEP amplitudes using an exponential fit on the fEP amplitudes over the first 5 stimulus pulses across diseases. (G) Shows the half-life of the fitted exponential (i.e., rate of attenuation of fEP amplitudes) between PD and dystonia, with PD demonstrating faster fEP attenuation.

Offline analyses and statistics

GPI high-frequency discharge (HFD) neurons [17] of >4 signal-to-noise ratio and <1% interspike interval violations were included in the single-neuron analyses. Segments were band-pass filtered (300-3000Hz) and template matched, after which neurophysiological features were extracted from each segment, including per patient median firing rate (FR), and the ninetieth percentile of the burst index (BI), coefficient of variation (CV), and oscillatory power across the theta (4-8Hz), alpha (8-12Hz), low-beta (12-21Hz), and high-beta (21-30Hz) frequency bands.

Oscillatory power was extracted using Lomb's periodogram, performed on the autocorrelation function of single-neuron segments; as previously described [19]. The burst index was computed by taking the ratio of the means from a two-component Gaussian mixture model applied to the log interspike interval distribution, a modification of the previous mode-over-mean ISI method [20]. Neuronal features were compared between PD and dystonia using 2-tailed Mann-Whitney U (MWU) tests due to non-normality of features. Spearman correlations were employed to investigate relationships between extracted features and clinical scores. For comparing differences in single-neuron features between PD and dystonia, significant results were followed up with post hoc multiple comparisons with a Bonferroni correction. For clinical correlations, non-parametric Monte Carlo permutation tests were used, avoiding assumptions about data distribution. The tested values were randomly shuffled 5,000 times to form a probability distribution, with the p-value reflecting the original sample rank. All tests underwent adjustment for multiple comparisons, controlling the false discovery rate (FDR) at an α -level of 0.05.

All fEP measurements for plasticity analyses were normalized with respect to the root mean square amplitude of the rectified pre-stimulation LFP (low-pass filtered at 50Hz). For long-term plasticity analyses, the median normalized fEP amplitudes pre-versus post-HFS at each recording site were analyzed using a linear mixed model (LMM) with patient ID as a random factor, normalized fEP amplitudes as the response variable, and epoch as a fixed effect, to establish whether plasticity was elicited in each disorder. The percentage change from pre- to post-HFS was also compared across disorders using a LMM with patient ID as a random factor. For short-term plasticity analysis, the dynamic change of the first five fEP amplitudes during the first train of HFS was assessed; normalized with respect to the first fEP amplitude for each train. Given the relatively rapid rate of attenuation of fEP amplitudes in GPI with HFS [16,18], the first five fEPs were fit with a double-exponential function, and the half-life of the faster decaying exponential component was used as a readout of the rate of attenuation. Half-life measurements were then compared across disorders using a LMM with patient ID as a random factor.

Data Sharing

The data that support the findings of this study are available in a public repository (see: <https://osf.io/nqzd2/>)

Results

Differences in single-neuron features

In dystonia, FR was lower (MWU test; $p=.0252$) and neurons were burstier (higher BI; MWU test; $p=.0024$) and less regular (higher CV; MWU test; $p<.0001$) compared to PD (**Fig. 1A**). However, there were no significant differences in spiketrain oscillations in the delta (MWU test; $p=.6265$),

theta (MWU test; $p=.3649$), low beta (MWU test; $p=.1176$), or high beta (MWU test; $p=.4585$) frequency bands (**Fig. 1B**). The post hoc Bonferroni correction (adjusted significance threshold to $p < 0.0071$) indicated that firing rate difference between PD and dystonia did not survive multiple comparisons (Supplementary Table 3). A neuronal feature summary is available in Supplementary Table 2.

Clinical correlations

For rate-based spiketrain features, no clinical correlations were found for PD. In dystonia, symptom severity negatively correlated with FR (Spearman's $\rho=-0.49$, $p=.0370$) and positively correlated with CV (Spearman's $\rho=0.48$, $p=.0364$; **Fig. 1C**). For oscillatory spiketrain features, symptom severity positively correlated with low beta in PD (Spearman's $\rho=0.30$, $p=.0426$), with a marginally stronger correlation for PD hypokinetic symptoms only (items 23-26 of UPDRSIII, Spearman's $\rho=0.32$, $p=.0330$; Supplementary Fig. 3), and theta in dystonia (Spearman's $\rho=0.51$, $p=.0242$; **Fig. 1C**). However, none of the clinical correlations survived Benjamini-Hochberg FDR-correction for multiple comparisons (Supplementary Table 4).

Long-term plasticity-like effects

In PD, fEP amplitudes were significantly greater after compared to before HFS (LMM; $p = .0075$, effect size = 5.42 ± 1.79 ; **Fig. 2C**), while in dystonia, the increase approached but did not reach statistical significance (LMM; $p = .0708$, effect size = 2.82 ± 1.45 ; **Fig. 2C**). The percent change in fEP amplitude was significantly greater in PD compared to dystonia (LMM; $p = .0396$, effect size = 12.40 ± 5.41 ; **Fig. 2D**).

Short-term plasticity

Half-life values extracted from double exponential fits on the first five fEPs during HFS were significantly greater in dystonia compared to PD (LMM; $p = 0.0139$, effect size = 1.04 ± 0.37 ; **Fig. 2G**).

Discussion

In this work, we leveraged the intraoperative environment to characterize GPi spiketrain features and to record and manipulate the efficacy of inhibitory direct pathway striato-pallidal projections, in order to derive insights about disease-related features of basal ganglia circuit function in PD and dystonia. In particular, we found that GPi neurons exhibited lower firing rates, but greater burstiness and variability in dystonia compared to PD (**Fig. 1A**). While no differences were found in the power of spiketrain oscillations across disorders (**Fig. 1B**), we found that PD symptom severity positively correlated with the power of low-beta frequency spiketrain oscillations, whereas dystonia symptom severity positively correlated with the power of theta frequency spiketrain oscillations (**Fig. 1C**). Dystonia symptom severity moreover correlated negatively with firing rate, and positively with neuronal variability. These results are discussed in greater detail with respect to previous literature in the subsequent Discussion section entitled “*Neuronal correlates of PD and dystonia*.” In response to electrical stimulation (protocol depicted in **Fig. 2A**), we found significant increases in the amplitudes of positive-going stimulation-evoked field potential amplitudes (considered to reflect striato-pallidal synaptic strength; as exemplified in **Fig. 2B**) before versus after HFS in both PD and dystonia (**Fig. 2C**); with recording sites in PD exhibiting significantly greater increases (**Fig. 2D**). While changes to evoked potential amplitude before versus after stimulation can be considered to be reflective of long-term plasticity [15,18], the dynamics of evoked potentials during HFS (as depicted in **Fig. 2E**) can be considered as reflective of short-term synaptic plasticity [18,21]. To this end, our findings are

suggestive of faster latency synaptic depression in PD compared to dystonia (**Fig. 2F** [G](#)). Plasticity findings are discussed in greater detail in the Discussion section entitled “*Direct pathway plasticity*.”

Neuronal correlates of PD and dystonia

The basal ganglia circuit models of PD and dystonia have been suggested to be driven by an imbalance between direct and indirect pathways [22 [G](#), 23 [G](#)]. While hyperfunctionality of the indirect pathway and hypofunctionality of the direct pathway are associated with PD [24 [G](#)], the direct pathway has been suggested to be hyperfunctional in dystonia [7 [G](#)]. To this end, our findings (**Fig. 1A** [G](#)) of lower firing rates in dystonia compared to PD may further substantiate [8 [G](#), 9 [G](#)] such claims, whereby greater inhibition of GPi would produce decreased inhibitory output, and thus under-inhibition of thalamocortical motor networks, giving rise to hyperkinetic motor symptoms; whereas the opposite would be true in PD [3 [G](#)]. These claims are further substantiated by findings of an inverse relationship between firing rate and disease severity in dystonia (**Fig. 1C** [G](#); rate-based features); although the opposite was not found in PD. Despite the lack of correlations with firing rate in PD, our findings seem to align with those of Muralidharan and colleagues [25 [G](#)], who showed that GPi neuronal firing rates may not directly correlate with motor severity but exhibit variability across the disease severity continuum in parkinsonian non-human primates (initially increasing, then decreasing, then increasing again at mild, moderate, and severe disease manifestations, respectively). Thus, while GPi discharge rates may change in PD, such changes may not be reflected by linear relationships with motor sign development and progression. Indeed, variability in spike firing rates in PD may be reflected in the considerable overlap in spiking activity between PD and dystonia (**Fig. 1A** [G](#)), with many dystonia patients exhibiting higher discharge rates compared to PD patients. While differences in discharge rates were nevertheless observed between PD and dystonia, it may be that the combination of rate and pattern (reflected in the BI and CV) changes best differentiates the two disorders.

From a neural synchronization standpoint, we recently showed that pathological oscillations at the neural aggregate / LFP level are likely encoded by periodic oscillations in the firing rates of single neurons [26 [G](#)]. In the current work, we did not find significant differences in the power of spiketrain oscillations across disorders (**Fig. 1B** [G](#)). Although previous research has reported differences in the LFP power between PD and dystonia [27 [G](#), 28 [G](#)], a study in non-human primates found no such differences in single-neuron oscillatory strength [8 [G](#)], as reflected in our findings. However, despite a lack of difference in overall power across disorders, we were able to derive disease/frequency-specific relationships with respect to clinical scores (**Fig. 1C** [G](#); oscillatory features). In particular, we provided evidence of relationships between the power of spiketrain oscillations in the theta and low beta frequency bands and the severity of motor symptoms in dystonia and PD, respectively; providing a cellular level validation of previous frequency-specific LFP clinical correlation studies in people with PD (beta oscillations) [11 [G](#)] and dystonia (theta oscillations) [12 [G](#)]. Indeed, Muralidharan and colleagues [25 [G](#)] also showed linear group-level relationships between low-beta frequency spiketrain oscillations and disease severity in parkinsonian non-human primates, despite the lack of linear relationships with spike discharge rates (as discussed above).

Direct pathway plasticity

We have previously shown that the amplitude of positive-going inhibitory fEPs is directly associated with the duration of neuronal inhibition [15 [G](#), 29 [G](#)], thus serving as a viable substrate of inhibitory synaptic transmission efficacy. These observations are supported by work in animals demonstrating the reversal of both positive-going extracellular fields and neuronal inhibition with the application of GABA antagonists [30 [G](#), 31 [G](#)]. Importantly, we previously [29 [G](#)] demonstrated that low latency inhibitory fields recorded in the subthalamic nucleus (GPe-mediated) are far smaller in amplitude and are resilient during HFS (i.e., minimal synaptic depression), whereas high latency inhibitory fields recorded in substantia nigra pars reticulata and GPi [16 [G](#)]

(striatum-mediated) are far larger in amplitude and are subject to potent synaptic depression during HFS; thus, enabling electrophysiological discernment of these distinct inhibitory projections in humans.

To this effect, our findings may suggest disease-specific differences in mechanisms underlying both short-[32] and long-term [33,34] forms of synaptic plasticity of direct pathway projections (Fig. 2). Converging evidence from past animal and human studies suggests that dystonia is associated with impaired synaptic function and abnormal synaptic plasticity [35–37]. Compared to healthy controls, it has been shown that transcranial magnetic stimulation induced motor evoked potentials (MEPs) are hyperexcitable in dystonia [38,39], and somatosensory and motor cortical plasticity is greater [40]. Likewise, enhanced long-term potentiation at cortico-striatal synapses has been shown in rodent models of dystonia [41,42]. While our finding that long-term potentiation effects are greater in PD compared to dystonia (Fig. 2D) is difficult to corroborate with this literature, one potential explanation can be that all of our PD patients are long-term users of levodopa. We have previously shown that the intake of this antiparkinsonian dopaminergic medication leads to potent increases in the magnitude of direct pathway plasticity [15]. Although patients are 12hr withdrawn from antiparkinsonian medications for surgery, it could be that striato-pallidal synapses are nevertheless chronically over-sensitized from prolonged use of dopaminergic medication; which is a well-known hypothesis related to the manifestation of levodopa-induced dyskinesias (a hyperkinetic feature) in PD [43]. Indeed, a lack of depotentiation of striato-pallidal projections has previously been observed in patients with levodopa-induced dyskinesias [44]. As such, excessive plasticity of these projections may corroborate hyperkinetic features of dystonia and levodopa-induced dyskinesias in PD.

In addition, the variation in the modulation of these striato-pallidal projections to electrical stimulation may also be reflective of differences in the mechanism by which DBS operates across PD and dystonia, despite a common stimulation target. Clinical studies in dystonia have shown that DBS leads to a more rapid improvement in the transient, dynamic muscle contractions (phasic components) of the disorder when compared to the sustained, continuous muscle contractions (tonic or fixed components) [45]. The early improvement of the phasic component has been suggested to be associated with direct suppression of abnormal activity in the GPi [46], whereas the improvement of tonic components has been associated with long-term potentiation-like synaptic plasticity in TMS studies [14,47]. This contrast with PD, where the maximal clinical response to DBS occurs within a much faster time course [13,48]. The present study shows disease-specific differences in the dynamics associated with modulation of direct pathway projections; thus, potentially providing a local synaptic rationale for the differential time courses associated with the clinical effects of DBS in the same target.

Specifically, we found faster rates of synaptic depression during HFS and greater post-HFS long-term plasticity-like effects in PD compared to dystonia, implying that striato-pallidal synapses in dystonia are perhaps more resistant to change compared to PD. Our findings may therefore explain the phenomenon that functional benefits of DBS require longer time courses in dystonia, if the effects of DBS rely upon network reorganization, which involves adjustments in neural connectivity or synaptic efficacy in response to stimulation [47]. Indeed, the benefits of GPi-DBS in dystonia have also been shown to persist for long periods of time even after the cessation of stimulation [14]. In PD, washout effects take longer with GPi-DBS compared to STN-DBS [49], and long-term dyskinesia reduction has also been observed with GPi-DBS even when DBS is off [50,51]. While further work is certainly required to better understand disease-related differences in plasticity, our findings may nevertheless motivate the development of periodic intermittent (ON/OFF) DBS strategies which periodically modulate synaptic plasticity for therapeutic benefits which outlast stimulation delivery, as have recently been employed in preclinical work [52,53].

Limitations and future directions

Important limitations of intraoperative/intracranial studies in humans include a lack of access to healthy control data (hence comparison across disorders), the inability to use pharmacological interventions to verify pathway specificity of elicited responses, and time constraints preventing thorough scrutinization of time courses of long-term plasticity-like effects. Indeed, GPi receives the greatest abundance of inhibitory inputs from striatum (direct pathway), but also it also receives inhibitory inputs by way of GPe (indirect pathway). Although we can functionally disaggregate these pathway-specific responses based on differences in morphology and dynamics of GPe-mediated versus striatum-mediated inhibitory fEPs [21]; the possibility of compounded effects cannot be completely ruled out. While our neuronal investigations provide cellular level support for closed-loop targeting of disease-related neural oscillations [54,55], future applications of DBS may also benefit from closed loop tuning of basal-ganglia-thalamo-cortical circuit dynamics and plasticity through chronic monitoring of evoked potential responses [56].

Indeed, optogenetic studies in parkinsonian rodents have demonstrated the ability to achieve lasting therapeutic efficacy via periodic activations of striatal direct pathway projections [52,53], likely leveraging long-term plasticity-like mechanisms. An additional limitation of this study is that we did not stringently monitor personal medication intake or levels of intraoperative sedation; however, all patients with PD were asked to withdraw from medications the night before surgery, and the analyses only included patients operated on awake. It is important to consider that classifications of PD and dystonia as hypo- and hyperkinetic disorders can be considered oversimplifications, as there can be contradictory comorbidities and drug- and DBS-related effects [57,58]; though our results do not change substantially when only hypokinetic PD features are considered (Supplementary Fig. 3). Additionally, our analyses involved pooling of patients with various forms of dystonia. While we did not observe differences across dystonia subtypes (Supplementary Fig. 1), future studies in larger patient cohorts would be warranted.

Finally, as many findings in Fig. 1 do not survive corrections for multiple comparisons, we suggest interpretation of results with caution. Despite this, many of our findings related to neuronal correlates are generally in line with previous literature, especially related to oscillatory correlates of PD and dystonia.

Conclusion

We substantiated claims of hypo-versus hyperfunctional GPi output in PD and dystonia, while furthermore providing cellular-level validation of the pathological nature of theta and low-beta oscillations in respective disorders. Such circuit changes may be underlain by disease-related differences in plasticity of striato-pallidal synapses, which are seemingly less plastic and/or respond slower to change in dystonia compared to PD.

Acknowledgements

The authors thank the patients for their participation.

Author Roles

(1) Research project: A. Conception, B. Organization, C. Execution; (2) Statistical analysis: A. Design, B. Execution, C. Review and critique; (3) Manuscript preparation: A. Writing of the first draft, B. Review and critique.

S.S: 1C, 2A, 2B, 3A; K.A.S: 1C, 2A, 2B, 3A; L.A.S: 1C, 2A, 2C, 3A; C.F: 1C, 3B; E.H: 1C, 3A; A.A.K: 1B, 3B; M.H: 1B, 3B; S.K.K: 1B, 3B; A.M.L: 1B, 3B; A.F: 1B, 3B; W.D.H: 1A, 1B, 1C, 2B, 3B; L.M: 1A, 1B, 1A, 1C, 2B, 3B.

References

- 1 Jankovic J. (2008) **Parkinson's disease: clinical features and diagnosis** *J Neurol Neurosurg Psychiatry* **79**:368–76
- 2 Albanese A, Bhatia K, Bressman SB, et al. (2013) **Phenomenology and classification of dystonia: A consensus update** *Mov Disord* **28**:863–73
- 3 DeLong MR, Wichmann T. (2007) **Circuits and Circuit Disorders of the Basal Ganglia** *Arch Neurol* **64**:20–4
- 4 DeLong MR (1990) **Primate models of movement disorders of basal ganglia origin** *Trends Neurosci* **13**:281–5
- 5 Naumann M, Pirker W, Reiners K, et al. (1998) **Imaging the pre- and postsynaptic side of striatal dopaminergic synapses in idiopathic cervical dystonia: A SPECT STUDY Using [123I] epidepride and [123I] β-CIT** *Mov Disord* **13**:319–23
- 6 Simonyan K, Berman BD, Herscovitch P, et al. (2013) **Abnormal Striatal Dopaminergic Neurotransmission during Rest and Task Production in Spasmodic Dysphonia** *J Neurosci* **33**:14705–14
- 7 Simonyan K, Cho H, Hamzehei Sichani A, et al. (2017) **The direct basal ganglia pathway is hyperfunctional in focal dystonia** *Brain* **140**:3179–90
- 8 Starr PA, Rau GM, Davis V, et al. (2005) **Spontaneous Pallidal Neuronal Activity in Human Dystonia: Comparison With Parkinson's Disease and Normal Macaque** *J Neurophysiol* **93**:3165–76
- 9 Tang JKH, Moro E, Mahant N, et al. (2007) **Neuronal firing rates and patterns in the globus pallidus internus of patients with cervical dystonia differ from those with Parkinson's disease** *J Neurophysiol* **98**:720–9
- 10 Hutchison WD, Lang AE, Dostrovsky JO, et al. (2003) **Pallidal neuronal activity: Implications for models of dystonia** *Ann Neurol* **53**:480–8
- 11 Neumann W-J, Degen K, Schneider G-H, et al. (2016) **Subthalamic synchronized oscillatory activity correlates with motor impairment in patients with Parkinson's disease** *Mov Disord* **31**:1748–51
- 12 Neumann W-J, Horn A, Ewert S, et al. (2017) **A localized pallidal physiomarker in cervical dystonia** *Ann Neurol* **82**:912–24
- 13 Stefani A, Cerroni R, Mazzone P, et al. (2019) **Mechanisms of action underlying the efficacy of deep brain stimulation of the subthalamic nucleus in Parkinson's disease: central role of disease severity** *Eur J Neurosci* **49**:805–16
- 14 Ruge D, Cif L, Limousin P, et al. (2011) **Shaping reversibility? Long-term deep brain stimulation in dystonia: the relationship between effects on electrophysiology and clinical symptoms** *Brain* **134**:2106–15

- 15 Milosevic L, Gramer R, Kim TH, et al. (2019) **Modulation of inhibitory plasticity in basal ganglia output nuclei of patients with Parkinson's disease** *Neurobiol Dis* **124**:46–56
- 16 Liu LD, Prescott IA, Dostrovsky JO, et al. (2012) **Frequency-dependent effects of electrical stimulation in the globus pallidus of dystonia patients** *J Neurophysiol* **108**:5–17
- 17 Hutchison WD, Lozano AM, Davis KD, et al. (1994) **Differential neuronal activity in segments of globus pallidus in Parkinson's disease patients** *Neuroreport* **5**:1533–7
- 18 Milosevic L, Kalia SK, Hodaie M, et al. (2018) **Neuronal inhibition and synaptic plasticity of basal ganglia neurons in Parkinson's disease** *Brain* **141**:177–90
- 19 Kaneoke Y, Vitek JL (1996) **Burst and oscillation as disparate neuronal properties** *J Neurosci Methods* **68**:211–23
- 20 Hutchison WD, Lozano AM, Davis KD, et al. (1994) **Differential neuronal activity in segments of globus pallidus in Parkinson's disease patients** *Neuroreport* **5**:1533–7
- 21 Steiner LA, Kühn AA, Geiger JRP, et al. (2022) **Persistent synaptic inhibition of the subthalamic nucleus by high frequency stimulation** *Brain Stimulat* **15**:1223–32
- 22 Hallett M. (1998) **The Neurophysiology of Dystonia** *Arch Neurol* **55**:601–3
- 23 Hallett M. (1993) **Physiology of Basal Ganglia Disorders: An Overview** *Can J Neurol Sci* **20**:177–83
- 24 McGregor MM, Nelson AB (2019) **Circuit Mechanisms of Parkinson's Disease** *Neuron* **101**:1042–56
- 25 Muralidharan A, Jensen AL, Connolly A, et al. (2016) **Physiological changes in the pallidum in a progressive model of Parkinson's disease: Are oscillations enough?** *Exp Neurol* **279**:187–96
- 26 Scherer M, Steiner LA, Kalia SK, et al. (2022) **Single-neuron bursts encode pathological oscillations in subcortical nuclei of patients with Parkinson's disease and essential tremor** *Proc Natl Acad Sci* **119**
- 27 Silberstein P, Kühn AA, Kupsch A, et al. (2003) **Patterning of globus pallidus local field potentials differs between Parkinson's disease and dystonia** *Brain J Neurol* **126**:2597–608
- 28 Wang DD, de Hemptinne C, Miocinovic S, et al. (2018) **Pallidal Deep-Brain Stimulation Disrupts Pallidal Beta Oscillations and Coherence with Primary Motor Cortex in Parkinson's Disease** *J Neurosci Off J Soc Neurosci* **38**:4556–68
- 29 Steiner LA, Kuehn AA, Geiger JR, et al. (2021) **Persistent synaptic inhibition of the subthalamic nucleus by high frequency stimulation** *bioRxiv* <https://doi.org/10.1101/2021.10.20.465131>
- 30 Yoshida M, Precht W. (1971) **Monosynaptic inhibition of neurons of the substantia nigra by caudatonigral fibers** *Brain Res* **32**:225–8

- 31 Precht W, Yoshida M. (1971) **Blockage of caudate-evoked inhibition of neurons in the substantia nigra by picrotoxin** *Brain Res* **32**:229–33
- 32 Prescott IA, Dostrovsky JO, Moro E, et al. (2013) **Reduced paired pulse depression in the basal ganglia of dystonia patients** *Neurobiol Dis* **51**:214–21
- 33 Ueki Y, Mima T, Kotb MA, et al. (2006) **Altered plasticity of the human motor cortex in Parkinson's disease** *Ann Neurol* **59**:60–71
- 34 Tamura Y, Ueki Y, Lin P, et al. (2009) **Disordered plasticity in the primary somatosensory cortex in focal hand dystonia** *Brain* **132**:749–55
- 35 Chen R, Udupa K. (2009) **Measurement and modulation of plasticity of the motor system in humans using transcranial magnetic stimulation** *Motor Control* **13**:442–53
- 36 Peterson DA, Sejnowski TJ, Poizner H. (2010) **Convergent evidence for abnormal striatal synaptic plasticity in dystonia** *Neurobiol Dis* **37**:558–73
- 37 Quartarone A, Pisani A. (2011) **Abnormal plasticity in dystonia: Disruption of synaptic homeostasis** *Neurobiol Dis* **42**:162–70
- 38 Chen R. (2000) **Studies of human motor physiology with transcranial magnetic stimulation** *Muscle Nerve* **23**:S26–32
- 39 Siebner HR, Tormos JM, Ceballos-Baumann AO, et al. (1999) **Low-frequency repetitive transcranial magnetic stimulation of the motor cortex in writer's cramp** *Neurology* **52**:529–37
- 40 Tamura Y, Ueki Y, Lin P, et al. (2009) **Disordered plasticity in the primary somatosensory cortex in focal hand dystonia** *Brain J Neurol* **132**:749–55
- 41 Köhling R, Koch U-R, Hamann M, et al. (2004) **Increased excitability in cortico-striatal synaptic pathway in a model of paroxysmal dystonia** *Neurobiol Dis* **16**:236–45
- 42 Martella G, Tassone A, Sciamanna G, et al. (2009) **Impairment of bidirectional synaptic plasticity in the striatum of a mouse model of DYT1 dystonia: role of endogenous acetylcholine** *Brain J Neurol* **132**:2336–49
- 43 Calabresi P, Ghiglieri V, Mazzocchetti P, et al. (2015) **Levodopa-induced plasticity: a double-edged sword in Parkinson's disease?** *Philos Trans R Soc B Biol Sci* **370** <https://doi.org/10.1098/rstb.2014.0184>
- 44 Prescott IA, Liu LD, Dostrovsky JO, et al. (2014) **Lack of depotentiation at basal ganglia output neurons in PD patients with levodopa-induced dyskinesia** *Neurobiol Dis* **71**:24–33
- 45 Krauss JK (2002) **Deep Brain Stimulation for Dystonia in Adults** *Stereotact Funct Neurosurg* **78**:168–82
- 46 Barow E, Neumann W-J, Brücke C, et al. (2014) **Deep brain stimulation suppresses pallidal low frequency activity in patients with phasic dystonic movements** *Brain* **137**:3012–24

- 47 Ruge D, Tisch S, Hariz MI, et al. (2011) **Deep brain stimulation effects in dystonia: Time course of electrophysiological changes in early treatment** *Mov Disord* **26**:1913–21
- 48 Hammond C, Bergman H, Brown P. (2007) **Pathological synchronization in Parkinson's disease: networks, models and treatments** *Trends Neurosci* **30**:357–64
- 49 Follett KA, Weaver FM, Stern M, et al. (2010) **Pallidal versus subthalamic deep-brain stimulation for Parkinson's disease** *N Engl J Med* **362**:2077–91
- 50 Anderson VC, Burchiel KJ, Hogarth P, et al. (2005) **Pallidal vs Subthalamic Nucleus Deep Brain Stimulation in Parkinson Disease** *Arch Neurol* **62**:554–60
- 51 Bejjani B-P, Dormont D, Pidoux B, et al. (2000) **Bilateral subthalamic stimulation for Parkinson's disease by using three-dimensional stereotactic magnetic resonance imaging and electrophysiological guidance** *J Neurosurg* **92**:615–25
- 52 Spix TA, Nanivadekar S, Toong N, et al. (2021) **Population-specific neuromodulation prolongs therapeutic benefits of deep brain stimulation** *Science* **374**:201–6
- 53 Mastro KJ, Zitelli KT, Willard AM, et al. (2017) **Cell-specific pallidal intervention induces long-lasting motor recovery in dopamine-depleted mice** *Nat Neurosci* **20**:815–23
- 54 Johnson V, Wilt R, Gilron R, et al. (2021) **Embedded adaptive deep brain stimulation for cervical dystonia controlled by motor cortex theta oscillations** *Exp Neurol* **345**
- 55 Little S, Beudel M, Zrinzo L, et al. (2016) **Bilateral adaptive deep brain stimulation is effective in Parkinson's disease** *J Neurol Neurosurg Psychiatry* **87**:717–21
- 56 Versantvoort EM, Dietz BE, Mugan D, et al. (2024) **Evoked compound action potential (ECAP)-controlled closed-loop spinal cord stimulation in an experimental model of neuropathic pain in rats** *Bioelectron Med* **10**
- 57 Reese R, Fasano A, Knudsen K, et al. (2014) **Full Parkinsonian Triad Induced by Pallidal High-Frequency Stimulation in Cervical Dystonia** *Mov Disord Clin Pract* **2**:99–101
- 58 Esposito M, Dubbioso R, Peluso S, et al. (2017) **Cervical dystonia patients display subclinical gait changes** *Parkinsonism Relat Disord* **43**:97–100

Editors

Reviewing Editor

Christian Büchel

University Medical Center Hamburg-Eppendorf, Hamburg, Germany

Senior Editor

Christian Büchel

University Medical Center Hamburg-Eppendorf, Hamburg, Germany

Reviewer #1 (Public Review):

Summary:

Sumarac et al investigate differences in globus pallidus internus (GPi) spike activity and short- and long-term plasticity of direct pathway projections in patients with Parkinson's disease (PD) and dystonia. Their main claims are that GPi neurons exhibit distinct characteristics in these two disorders, with PD associated with specific power-frequency oscillations and dystonia showing lower firing rates, increased burstiness, and less regular activity. Additionally, long-term plasticity and synaptic depression appear to differ between the two conditions. The authors suggest that these findings support the concept of hyperfunctional GPi output in PD and hypofunctional output in dystonia, possibly driven by variations in plasticity of striato-pallidal synapses. Overall enthusiasm is relatively high, but I think the discussion omits discussing findings that don't align well with standard models.

Strengths:

- These types of studies are valuable as the data arise from patients who have dystonia or PD. This could provide unique insights into disease pathophysiology that might not be recapitulated in animal systems work.

Comments on latest version:

The authors addressed my concerns in their revision.

<https://doi.org/10.7554/eLife.90454.2.sa2>

Reviewer #2 (Public Review):

Summary:

The authors investigated how neuronal activity and metrics of plasticity using local electrical stimulation in the GPi were different between Parkinson's disease and dystonia patients.

Strengths:

The authors achieved their aims of comparing the dynamics related to stimulation induced metrics of plasticity in GPi between dystonia and PD, which has not been previously explored. These results could directly inform DBS protocols to improve treatment. The methods are clearly described and results are strong with measurements from a large population of patients for each disease group, and with distinct findings for each group. These results also may help provide insight as to the differences in terms of dynamics of therapeutic stimulation effects in the different disease groups.

Weaknesses:

After the revisions, the discussion contains many more details and comparisons to relevant literature, which will be helpful for readers to appreciate the importance of the results. The conclusion could have been strengthened as well, as it seems to be a very general summary of their findings without consideration of their clinical implications and importance. However, this may be a minor issue.

<https://doi.org/10.7554/eLife.90454.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment:

This manuscript is a valuable study of the responses of GPi neurons to DBS stimulation in human PD and dystonia patients and it finds evidence for altered short-term and long-term plasticity in response to DBS between the two patient populations. This data set is of interest to both basic and clinical researchers working in the field of DBS and movement disorders. While there was enthusiasm for the potential significance of these findings, support for their conclusions was incomplete. Their data may be indicative of more interesting and complex interpretations than currently considered in the article.

The authors would like to express their gratitude to the Editorial Team and Reviewers for their invaluable feedback which helped to improve the manuscript.

Reviewer #1:*Summary:*

Sumarac et al investigate differences in globus pallidus internus (GPi) spike activity and short- and long-term plasticity of direct pathway projections in patients with Parkinson's disease (PD) and dystonia. Their main claims are that GPi neurons exhibit distinct characteristics in these two disorders, with PD associated with specific power-frequency oscillations and dystonia showing lower firing rates, increased burstiness, and less regular activity. Additionally, long-term plasticity and synaptic depression appear to differ between the two conditions. The authors suggest that these findings support the concept of hyperfunctional GPi output in PD and hypofunctional output in dystonia, possibly driven by variations in the plasticity of striato-pallidal synapses. Overall enthusiasm is relatively high, but I think the discussion omits discussing findings that don't align well with standard models.

Strengths:

These types of studies are valuable as the data arise from patients who have dystonia or PD. This could provide unique insights into disease pathophysiology that might not be recapitulated in animal systems work.

Thank you for the positive feedback.

Weaknesses:

- The rate model and indirect/direct pathway ideas lack explanatory power; too much of the hypothesis generation and discussion in this manuscript is set in the context of these old ideas. Their data in my view emphasize this somewhat emphatically. Most patients with the 'hypokinetic' movement disorder PD have dystonia as a part of their motor features. Dystonia is a form of excessive muscle activation that on the one hand is 'hyperkinetic' but on the other usually decreases the speed of motor tasks, even in patients with primary dystonia. Similarly, PD patients display a bewildering variety of hyperkinetic manifestations as well (rest tremor, dystonia, dyskinesia). If these are truly independent classifications, i.e. hyper- versus hypo-kinetic, the authors must acknowledge that there is considerable overlap in the spike activity across groups - numerous dystonia patients display higher discharge rates than the majority of the PD sample. Based on the firing rate alone, it would not be possible to distinguish these groups.

Thank you for your insightful comments regarding the discussion of the rate model and the distinction between hyperkinetic and hypokinetic movement disorders. We acknowledge that

the rate model, primarily derived from limited number of animal subjects [1], may not fully encapsulate the complexities of Parkinson's disease (PD) and dystonia. Our study aimed to validate animal model findings in humans by correlating single-neuron features with disease symptom severity. However, we concur with the Reviewer's comment regarding the overlapping motor features in hypokinetic and hyperkinetic disorders. We can speculate that the overlap in neuronal properties may be reflected in the overlap of, for example, hyperkinetic features being also present in PD, as suggested by the Reviewer. Per the Reviewer's request, we have now acknowledged this notion in the manuscript. Interestingly, hypokinetic symptoms have been reported to occur in dystonia in response to GPi-stimulation and have been associated with beta activity in the LFP [2], which reinforces the notion that neural activity may be more related to specific symptoms rather than diseases as a whole. Supplementing our analyses, in addition to total UPDRSIII scores, we have now provided correlations with only hypokinetic (i.e. bradykinesia) subscores of the UPDRSIII to focus on more direct assessment of hypokinetic features in PD versus hyperkinetic features in dystonia. We have updated our methods and results accordingly.

[1] M. R. DeLong, "Primate models of movement disorders of basal ganglia origin.," Trends Neurosci, vol. 13, no. 7, pp. 281–285, Jul. 1990, doi: 10.1016/0166-2236(90)90110-v.

[2] R. Lofredi et al., "Pallidal Beta Activity Is Linked to Stimulation-Induced Slowness in Dystonia," Movement Disorders, vol. 38, no. 5, pp. 894–899, 2023, doi: 10.1002/mds.29347.

Amendments to the manuscript:

"Indeed, variability in spike firing rates in PD may be reflected in the considerable overlap in spiking activity between PD and dystonia (Fig. 1A), with many dystonia patients exhibiting higher discharge rates compared to PD patients."

"Given that UPDRSIII includes both hypokinetic and hyperkinetic symptoms of PD, we further sought to disaggregate the score by only considering items 23-26 in UPDRSIII, which assess hypokinetic symptoms of PD."

"... with a marginally stronger correlation for PD hypokinetic symptoms only (items 23-26 of UPDRSIII, Spearman's $\rho=0.32$, $p=.0330$; Supplementary Fig. 3)"

Supplementary Fig. 3: We provided correlations with hypokinetic (i.e., bradykinesia) subscore of the UPDRSIII. There is very little difference between correlation results of UPDRSIII total (Fig. 1) and the hypokinetic-only subscore (Supplementary Fig. 3).

"though our results do not change substantially when only hypokinetic PD features are considered (Supplementary Fig. 3)."

- If beta power is pathognomonic of parkinsonism, the authors found no differences in beta-related spike discharges across the groups. One would have predicted greater beta power in PD than in primary dystonia. This should be discussed explicitly and an interpretation should be provided.

We agree with the reviewer that considering the previous LFP literature, one might have expected a difference in single-neuron oscillation power between PD and dystonia. However, while prior studies [3], [4] have reported significant differences in oscillatory power between the two diseases, researchers examined local field potential (LFP) activity only. Other work [5] in non-human primates investigated single-neuron oscillations and reported no differences between PD and dystonia at the single-neuron level, in line with our findings. However, despite the lack of difference in overall power presented here, we provide evidence that the strength of the beta-frequency single-neuron oscillations nevertheless correlates with symptom severity in PD but not dystonia; whereas the strength of the theta-frequency single-neuron oscillations correlates with symptom severity in dystonia but not PD.

- [3] P. Silberstein et al., “Patterning of globus pallidus local field potentials differs between Parkinson’s disease and dystonia,” *Brain*, vol. 126, no. Pt 12, pp. 2597–2608, Dec. 2003, doi: 10.1093/brain/awg267.
- [4] D. D. Wang et al., “Pallidal Deep-Brain Stimulation Disrupts Pallidal Beta Oscillations and Coherence with Primary Motor Cortex in Parkinson’s Disease,” *J Neurosci*, vol. 38, no. 19, pp. 4556–4568, May 2018, doi: 10.1523/JNEUROSCI.0431-18.2018.
- [5] P. A. Starr et al., “Spontaneous pallidal neuronal activity in human dystonia: comparison with Parkinson’s disease and normal macaque,” *J Neurophysiol*, vol. 93, no. 6, pp. 3165–3176, Jun. 2005, doi: 10.1152/jn.00971.2004.

Amendments to the manuscript:

“Although previous research has reported differences in the LFP power between PD and dystonia [27,28], a study in non-human primates found no such differences in single-neuron oscillatory strength [8], as reflected in our findings. However, despite a lack of difference in overall power across disorders, we were able to derive disease/frequency-specific relationships with respect to clinical scores (Fig. 1C; oscillatory features).”

- The study lacks a healthy control group, making it challenging to differentiate disease-specific findings from normal variations in GPi activity and plasticity. Although this is acknowledged in the discussion, this complicates the interpretation of the results. The sample sizes for PD and dystonia patients are relatively small, and the study combines various forms of dystonia, potentially masking subtype-specific differences. A larger and more homogenous sample could enhance the study's reliability.

Indeed, intraoperative microelectrode recordings cannot be obtained in healthy individuals. We agree with the Reviewer that this limits the interpretation of the data. However, directly comparing clinical correlations with single neuron readouts between two distinct clinical entities may, to some degree, compensate for the lack of healthy control data. This contrast, while not providing a healthy control, is still able to point to disease-specific differences. This approach has previously been used to comparisons at the LFP level [6]. While the sample size is indeed small, it is comparable or even higher to similar studies that have investigated the relation of symptom severity of single neuron readouts [7]. The Reviewer is right in that we do not differentiate between generalized or cervical dystonia. We chose to do so because our subgroup analysis provided in the Supplementary Material did not suggest specific differences; though there is insufficient data from specific dystonia subtypes to make formal statistical comparisons. Indeed, future studies should investigate specific subtypes further.

- [6] R. Lofredi et al., “Pallidal beta bursts in Parkinson’s disease and dystonia,” *Movement Disorders*, vol. 34, no. 3, pp. 420–424, 2019, doi: 10.1002/mds.27524.

- [7] A. Gulberti et al., “Subthalamic and nigral neurons are differentially modulated during parkinsonian gait,” *Brain*, p. awad006, Feb. 2023, doi: 10.1093/brain/awad006.

Amendments to the manuscript:

“While we did not observe differences across dystonia subtypes (Supplementary Fig. 1), future studies in larger patient cohorts would be warranted. Finally, as many findings in Fig. 1 do not survive corrections for multiple comparisons, we suggest interpretation of results with caution. Despite this, many of our findings related to neuronal correlates are generally in line with previous literature, especially related to oscillatory correlates of PD and dystonia.”

- While they mention that data are available on request, sharing data openly would increase transparency and allow for independent validation of the results. It is unclear how sharing deidentified data would compromise patient privacy or present ethical issues of any kind, as claimed by the authors.

Much of the data in question were collected under an old Research Ethics Board (REB) protocol which did not address data sharing. However, we have consulted with our REB and gained retroactive permission to post de-identified data which are now available in the Supplementary Material.

Amendments to the manuscript:

“The data that support the findings of this study are available in a public repository (see: <https://osf.io/nqzd2/>)”

- They appropriately acknowledge several limitations, such as the inability to use pharmacological interventions and the need for further research in the chronic setting.

Thank you for the comment.

- The manuscript highlights differences in GPi activity and plasticity between PD and dystonia but could provide more context on the clinical implications of these findings, particularly regarding what the implications would be novel paradigms for deep brain stimulation.

Thank you for the comment. Our finding that striato-pallidal plasticity decays more slowly in dystonia compared to PD may relate to the slower time course of symptom relief associated with GPi-DBS in dystonia, as presently outlined in the discussion. On the other hand, symptoms are also suppressed for longer after the cessation of stimulation in dystonia compared to PD, which may reflect long-term plastic changes [8], [9]. In the context of clinical DBS, plasticity modulation may be facilitated by intermittent stimulation algorithms that may achieve the necessary plastic network change by applying stimulation for a defined time but could then be switched off for improved energy consumption and perhaps as a means of mitigating side effects. DBS devices with chronic sensing may enable monitoring of evoked potential amplitudes for future adaptive stimulation applications; however, currently available devices are limited by low sampling rates, but future devices may overcome these technical limitations.

[8] D. Ruge et al., “Deep brain stimulation effects in dystonia: time course of electrophysiological changes in early treatment,” *Mov Disord*, vol. 26, no. 10, pp. 1913–1921, Aug. 2011, doi: 10.1002/mds.23731.

[9] D. Ruge et al., “Shaping reversibility? Long-term deep brain stimulation in dystonia: the relationship between effects on electrophysiology and clinical symptoms,” *Brain*, vol. 134, no. Pt 7, pp. 2106–2115, Jul. 2011, doi: 10.1093/brain/awr122.

Amendments to the manuscript:

“While further work is certainly required to better understand disease-related differences in plasticity, our findings may nevertheless motivate the development of periodic intermittent (ON/OFF) DBS strategies which periodically modulate synaptic plasticity for therapeutic benefits which outlast stimulation delivery, as have recently been employed in preclinical work [52,53].”

- While statistical tests are mentioned, the manuscript could benefit from a more detailed presentation of statistical methods, including correction for multiple comparisons and effect sizes. Did the authors consider different recording sites within each patient as independent observations? I think this is not appropriate if that was the case.

Thank you for your constructive feedback. In response to the concerns regarding the statistical methods, we have expanded our analysis to provide a more comprehensive statistical overview. Specifically, we implemented the Bonferroni correction for multiple comparisons across each of the seven tests conducted for the differences in single-neuron features between PD and dystonia. The adjustment revealed that only the burst index and coefficient of variation retain statistical significance after post hoc correction, while the firing rate does not. Results of the Bonferroni corrections are now presented in Supplementary Table 3. Reflecting on the initial comment about firing rates between the two disorders, our updated findings underscore the limitation of using firing rates alone to differentiate between PD and dystonia, and instead, our analysis now points to burstiness and firing irregularity as more reliable discriminators. Regarding the clinical correlations, we refined our statistical analysis by employing nonparametric Monte Carlo permutation tests with 5000 permutations, as used in recent work [10], [11]. This method is chosen for its independence from assumptions regarding data distribution. Specifically, we computed and tested the Spearman rho for significance using the permutation test. Then, to address multiple comparisons, we controlled the false discovery rate (FDR) using the Benjamini-Hochberg procedure. Results of these comparisons are now presented in Supplementary Table 4. Lastly, to address the concern regarding recording site independence within patients, we updated our plasticity analysis methodology. In our study, 6 out of 18 patients had multiple recording sites. Thus, to account for this, we employed linear mixed models (LMM) with patient ID as a random factor to appropriately account for the non-independence of these observations.

[10] v Lofredi et al., “Dopamine-dependent scaling of subthalamic gamma bursts with movement velocity in patients with Parkinson’s disease,” *Elife*, vol. 7, p. e31895, Feb. 2018, doi: 10.7554/eLife.31895.

[11] R. Lofredi et al., “Subthalamic beta bursts correlate with dopamine-dependent motor symptoms in 106 Parkinson’s patients,” *npj Parkinsons Dis.*, vol. 9, no. 1, Art. no. 1, Jan. 2023, doi: 10.1038/s41531-022-00443-3.

Amendments to the manuscript:

“For comparing differences in single-neuron features between PD and dystonia, significant results were followed up with post hoc multiple comparisons with a Bonferroni correction. For clinical correlations, non-parametric Monte Carlo permutation tests were used, avoiding assumptions about data distribution. The tested values were randomly shuffled 5,000 times to form a probability distribution, with the p-value reflecting the original sample rank. All tests underwent adjustment for multiple comparisons, controlling the false discovery rate (FDR) at an α -level of 0.05.”

“analyzed using a linear mixed model (LMM) with patient ID as a random factor, normalized fEP amplitudes as the response variable, and epoch as a fixed effect”

“using a LMM with patient ID as a random factor”

“However, none of the clinical correlations survived Benjamini-Hochberg FDR-correction for multiple comparisons (Supplementary Table 4).”

“In PD, fEP amplitudes were significantly greater after compared to before HFS (LMM; $p = .0075$, effect size = 5.42 ± 1.79 ; Fig. 2C), while in dystonia, the increase approached but did not reach statistical significance (LMM; $p = .0708$, effect size = 2.82 ± 1.45 ; Fig. 2C).”

All statistics were updated in the results section and the figures.

“Finally, as many findings in Fig. 1 do not survive corrections for multiple comparisons, we suggest interpretation of results with caution. Despite this, many of our findings related to neuronal correlates are generally in line with previous literature, especially related to oscillatory correlates of PD and dystonia.”

- *The manuscript could elaborate on the potential mechanisms underlying the observed differences in GPi activity and plasticity and their relevance to the pathophysiology of PD and dystonia.*

Thank you for your feedback. We have enhanced the manuscript by integrating additional discussions on previous studies related to plasticity in dystonia and PD (e.g., [12], [13]), which highlight excessive plasticity in dystonia. Although these may appear contradictory to our findings of increased plasticity in PD compared to dystonia, we propose (also justified by previous literature) that chronic dopaminergic medication use may lead to synaptic over-sensitization, which has been hypothesized as a biological mechanism underlying levodopa-induced dyskinesias (a hyperkinetic feature) in PD [14].

[12] Y. Tamura et al., “Disordered plasticity in the primary somatosensory cortex in focal hand dystonia,” *Brain*, vol. 132, no. Pt 3, pp. 749–755, Mar. 2009, doi: 10.1093/brain/awn348.

[13] D. A. Peterson, T. J. Sejnowski, and H. Poizner, “Convergent evidence for abnormal striatal synaptic plasticity in dystonia,” *Neurobiol Dis*, vol. 37, no. 3, pp. 558–573, Mar. 2010, doi: 10.1016/j.nbd.2009.12.003.

[14] P. Calabresi, B. Picconi, A. Tozzi, V. Ghiglieri, and M. Di Filippo, “Direct and indirect pathways of basal ganglia: a critical reappraisal,” *Nat Neurosci*, vol. 17, no. 8, pp. 1022–1030, Aug. 2014, doi: 10.1038/nn.3743.

Amendments to the manuscript:

“Converging evidence from past animal and human studies suggests that dystonia is associated with impaired synaptic function and abnormal synaptic plasticity [35–37]. Compared to healthy controls, it has been shown that transcranial magnetic stimulation induced motor evoked potentials (MEPs) are hyperexcitable in dystonia [38,39], and somatosensory and motor cortical plasticity is greater [40]. Likewise, enhanced long-term potentiation at cortico-striatal synapses has been shown in rodent models of dystonia [41,42]. While our finding that long term potentiation effects are greater in PD compared to dystonia (Fig. 2D) is difficult to corroborate with this literature, one potential explanation can be that all of our PD patients are long-term users of levodopa. We have previously shown that the intake of this antiparkinsonian dopaminergic medication leads to potent increases in the magnitude of direct pathway plasticity [15]. Although patients are 12hr withdrawn from antiparkinsonian medications for surgery, it could be that striato-pallidal synapses are nevertheless chronically over-sensitized from prolonged use of dopaminergic medication; which is a well-known hypothesis related to the manifestation of levodopa-induced dyskinesias (a hyperkinetic feature) in PD [43]. Indeed, a lack of depotentiation of striato-pallidal projections has previously been observed in patients with levodopa-induced dyskinesias [44]. As such, excessive plasticity of these projections may corroborate hyperkinetic features of dystonia and levodopa-induced dyskinesias in PD.”

Reviewer #2:

Summary:

The authors investigated how neuronal activity and metrics of plasticity using local electrical stimulation in the GPi were different between Parkinson's disease and dystonia patients.

Strengths:

The introduction highlights the importance of the work and the fundamental background needed to understand the rest of the paper. It also clearly lays out the novelty (i.e., that the dynamics of plastic effects in GPi between dystonia and PD have not been directly compared).

The methods are clearly described and the results are well organized in the figures.

The results are strong with measurements from a large population of patients for each disease group and with distinct findings for each group.

Thank you for the kind appraisal.

Weaknesses:

The discussion was hard to follow in several places, making it difficult to fully appreciate how well the authors' claims and conclusions are justified by their data, mostly in relation to the plasticity results. It may help to summarize the relevant findings for each section first and then further expand on the interpretation, comparison with prior work, and broader significance. Currently, it is hard to follow each section without knowing which results are being discussed until the very end of the section. With the current wording in the "Neuronal correlates.." section, it is not always clear which results are from the current manuscript, and where the authors are referring to past work.

Thank you for this feedback. The main findings are now summarized in a paragraph at the beginning of the Discussion section, before being discussed in comparison to other studies in the literature in subsequent sub-sections. Moreover, throughout the Discussion, findings from our study are now always reflected by a reference to the relevant figure to more easily differentiate current findings from previous literature. Additionally, Discussion sub-sections have been expanded to consider additional literature in response to various comments throughout the Review process (including the subsequent Review comment).

Amendments to the manuscript:

Paper findings are referenced to figures which depict the results at hand; discussion sub-sections expanded; and the following text has been added at the start of the Discussion:

“In particular, we found that GPi neurons exhibited lower firing rates, but greater burstiness and variability in dystonia compared to PD (Fig. 1A). While no differences were found in the power of spiketrain oscillations across disorders (Fig. 1B), we found that PD symptom severity positively correlated with the power of low-beta frequency spiketrain oscillations, whereas dystonia symptom severity positively correlated with the power of theta frequency spiketrain oscillations (Fig. 1C). Dystonia symptom severity moreover correlated negatively with firing rate, and positively with neuronal variability. These results are discussed in greater detail with respect to previous literature in the subsequent Discussion section entitled “Neuronal correlates of PD and dystonia.” In response to electrical stimulation (protocol depicted in Fig. 2A), we found significant increases in the amplitudes of positive-going stimulation-evoked field potential amplitudes (considered to reflect striato-pallidal synaptic strength; as exemplified in Fig. 2B) before versus after HFS in both PD and dystonia (Fig. 2C); with recording sites in PD exhibiting significantly greater increases (Fig. 2D). While changes to evoked potential amplitude before versus after stimulation can be considered to be reflective

of long-term plasticity [15,18], the dynamics of evoked potentials during HFS (as depicted in Fig. 2E) can be considered as reflective of short-term synaptic plasticity [18,21]. To this end, our findings are suggestive of faster latency synaptic depression in PD compared to dystonia (Fig. 2F/G). Plasticity findings are discussed in greater detail in the Discussion section entitled “Direct pathway plasticity.”

Also, I felt that more discussion could be used to highlight the significance of the current results by comparing and/or contrasting them to prior relevant work and mechanisms. The novelty or impact is not very clear as written. Could this be further substantiated in the Discussion?

Thank you for the feedback. The discussion has been expanded to include additional literature that is relevant to the findings reported in the manuscript. For example, with regards to the neuronal correlates sub-section, we now highlight the important findings [15] that show changes to the discharge rates and oscillatory tendencies of GPi neurons in non-human primates in response to staged MPTP applications to progressively titrate motor severity; these results substantiate our lack of correlation with firing rates in PD, and presence of a clinical correlation with beta oscillations. We additionally now emphasize human studies that found LFP power difference between PD and dystonia [3], [4]; but simultaneously highlight studies that did not find such differences in spike-train oscillations (in non-human primates) [5], which is reflective of our own findings. With regards to our plasticity sub-section, we have added new content related to previous literature on plasticity in dystonia and PD (also addressed in response to a query from Reviewer #1). For example, we bring to light a variety of previous studies [12], [13] emphasizing excessive plasticity in dystonia. However, while such studies may seem to contradict our findings of greater plasticity in PD compared to dystonia, we additionally provide hypotheses (justified by previous literature) that prolonged used of dopaminergic medication may result in synaptic over-sensitization, thus giving rise to levodopa-induced dyskinesias (a hyperkinetic feature) in PD [14].

[3] P. Silberstein et al., “Patterning of globus pallidus local field potentials differs between Parkinson’s disease and dystonia,” *Brain*, vol. 126, no. Pt 12, pp. 2597–2608, Dec. 2003, doi: 10.1093/brain/awg267.

[4] D. D. Wang et al., “Pallidal Deep-Brain Stimulation Disrupts Pallidal Beta Oscillations and Coherence with Primary Motor Cortex in Parkinson’s Disease,” *J Neurosci*, vol. 38, no. 19, pp. 4556–4568, May 2018, doi: 10.1523/JNEUROSCI.0431-18.2018.

[5] P. A. Starr et al., “Spontaneous pallidal neuronal activity in human dystonia: comparison with Parkinson’s disease and normal macaque,” *J Neurophysiol*, vol. 93, no. 6, pp. 3165–3176, Jun. 2005, doi: 10.1152/jn.00971.2004.

[12] Y. Tamura et al., “Disordered plasticity in the primary somatosensory cortex in focal hand dystonia,” *Brain*, vol. 132, no. Pt 3, pp. 749–755, Mar. 2009, doi: 10.1093/brain/awn348.

[13] D. A. Peterson, T. J. Sejnowski, and H. Poizner, “Convergent evidence for abnormal striatal synaptic plasticity in dystonia,” *Neurobiol Dis*, vol. 37, no. 3, pp. 558–573, Mar. 2010, doi: 10.1016/j.nbd.2009.12.003.

[14] P. Calabresi, B. Picconi, A. Tozzi, V. Ghiglieri, and M. Di Filippo, “Direct and indirect pathways of basal ganglia: a critical reappraisal,” *Nat Neurosci*, vol. 17, no. 8, pp. 1022–1030, Aug. 2014, doi: 10.1038/nn.3743.

[15] A. Muralidharan et al., “Physiological changes in the pallidum in a progressive model of Parkinson’s disease: Are oscillations enough?,” *Exp Neurol*, vol. 279, pp. 187–196, May 2016, doi: 10.1016/j.expneurol.2016.03.002.

Amendments to the manuscript:

“Despite the lack of correlations with firing rate in PD, our findings seem to align with those of Muralidharan and colleagues [25], who showed that GPi neuronal firing rates may not directly correlate with motor severity but exhibit variability across the disease severity continuum in parkinsonian non-human primates (initially increasing, then decreasing, then increasing again at mild, moderate, and severe disease manifestations, respectively). Thus, while GPi discharge rates may change in PD, such changes may not be reflected by linear relationships with motor sign development and progression. Indeed, variability in spike firing rates in PD may be reflected in the considerable overlap in spiking activity between PD and dystonia (Fig. 1A), with many dystonia patients exhibiting higher discharge rates compared to PD patients. While differences in discharge rates were nevertheless observed between PD and dystonia, it may be that the combination of rate and pattern (reflected in the BI and CV) changes best differentiates the two disorders.”

“Converging evidence from past animal and human studies suggests that dystonia is associated with impaired synaptic function and abnormal synaptic plasticity [35–37]. Compared to healthy controls, it has been shown that transcranial magnetic stimulation induced motor evoked potentials (MEPs) are hyperexcitable in dystonia [38,39], and somatosensory and motor cortical plasticity is greater [40]. Likewise, enhanced long-term potentiation (LTP) at cortico-striatal synapses has been shown in rodent models of dystonia [41,42]. While our finding that LTP effects are greater in PD compared to dystonia (Fig. 2D) is difficult to corroborate with this literature, one potential explanation can be that all of our PD patients are long-term users of levodopa. We have previously shown that the intake of this antiparkinsonian dopaminergic medication leads to potent increases in the amount of plasticity elicited in GPi [15]. Although patients are 12hr withdrawn from antiparkinsonian medications for surgery, it could be that striato-pallidal synapses are nevertheless chronically over-sensitized from prolonged use of dopaminergic medication; which is a well-known hypothesis related to the manifestation of levodopa-induced dyskinesias (a hyperkinetic feature) in PD [43]. Indeed, a lack of depotentiation of striato-pallidal projections has previously been observed in patients with levodopa-induced dyskinesias [44]. As such, excessive plasticity of these projections may corroborate hyperkinetic features of dystonia and levodopa-induced dyskinesias in PD.”

Some specific comments and questions about the Discussion:

Lines 209-211 - This sentence was hard to understand, could it be clarified?

Lines 211-213 - What do phasic and tonic components mean exactly? Could this be specifically defined? Are there specific timescales (as referred to in Intro)?

Lines 215-217 - It's not clear what was delayed in dystonia, and how the authors are trying to contrast this with the faster time course in PD. I think some of this is explained in the introduction, but could also be re-summarized here as relevant to the results discussed.

Lines 223-224 - I'm not sure I follow the implication that network reorganization leads to delayed functional benefits. Could this be further elaborated?

Reply & Amendments to the manuscript: Thank you for your feedback. We've made the following concise revisions to address the comments:

We've clarified lines 209-211 to explain that variations in electrical stimulation effects on pathways in PD and dystonia may reveal the operational mechanisms of DBS, despite a common target:

“The variation in the modulation of these projections / pathways to electrical stimulation may also indicate the mechanism by which DBS operates across PD and dystonia, despite a common stimulation target.”

In response to the second comment on lines 211-213 about phasic and tonic components, we now specify that phasic refers to dynamic muscle contractions, and tonic to continuous muscle contractions, providing clear definitions relevant to our context:

“Clinical studies in dystonia have shown that DBS leads to a more rapid improvement in the transient, dynamic muscle contractions (phasic components) of the disorder when compared to the sustained, continuous muscle contractions (tonic or fixed components) [33]”

For lines 215-217, we’ve refined our discussion to clearly contrast the delayed response in dystonia with the faster onset in PD:

“This contrast with PD, where the, the maximal clinical response to DBS occurs within a much faster time course [13,36].”

On lines 223-224, we’ve expanded the explanation of how network reorganization may lead to delayed functional benefits, highlighting adjustments in neural connectivity and synaptic efficacy in response to stimulation:

“which involves adjustments in neural connectivity or synaptic efficacy in response to the stimulation [14,35].”

| *Could the absence of a relationship between FR and disease in PD be discussed?*

Thank you for raising this point. Despite observing higher firing rates in PD compared to dystonia, it is unexpected that these rates do not correlate with symptom severity according to the rate model of PD [1]. However, despite the lack of correlations with firing rates, our findings align with similar animal work of Muralidharan et al. [15], which reported that neuronal firing rates within the GPi of rhesus monkeys did not increase linearly with respect to varying intensities of parkinsonian motor severity. We did however show that low beta oscillatory strength within the GPi may play a significant role in the manifestation of motor symptoms in PD; which is also in line with findings of Muralidharan and colleagues. As per the Reviewer’s request, we have included this content into our discussion.

[1] M. R. DeLong, “Primate models of movement disorders of basal ganglia origin,,” Trends Neurosci, vol. 13, no. 7, pp. 281–285, Jul. 1990, doi: 10.1016/0166-2236(90)90110-v.

[15] A. Muralidharan et al., “Physiological changes in the pallidum in a progressive model of Parkinson’s disease: Are oscillations enough?,” Exp Neurol, vol. 279, pp. 187–196, May 2016, doi: 10.1016/j.expneurol.2016.03.002.

Amendments to the manuscript:

“Despite the lack of correlations with firing rate in PD, our findings seem to align with those of Muralidharan and colleagues [25], who showed that GPi neuronal firing rates may not directly correlate with motor severity but exhibit variability across the disease severity continuum in parkinsonian non-human primates (initially increasing, then decreasing, then increasing again at mild, moderate, and severe disease manifestations, respectively). Thus, while GPi discharge rates may change in PD, such changes may not be reflected by linear relationships with motor sign development and progression.”

“Indeed, Muralidharan and colleagues [25] also showed linear group-level relationships between low-beta frequency spiketrain oscillations and disease severity in parkinsonian non-

human primates, despite the lack of linear relationships with spike discharge rates (as discussed above)."

It wasn't very clear how the direct pathway can be attributed to plasticity changes if the GPi makes up both the direct and indirect pathways. Could this be further clarified?

The reviewer brings up an important nuanced point. Recent work from our lab [16] shows that inhibitory evoked fields in STN (which receives inhibitory fields from GPe; no other inhibitory sources) are persistent with very minimal depression during HFS. On the other hand, inhibitory fields in the SNr (which receives majority of its inhibitory inputs from striatum; though some come by way of GPe as well per anatomical literature) depress quickly. We have previously also shown these rapidly depressing fields in GPi [17], [18], which also receives the majority of its inhibitory inputs via striatum, though some also from GPe. As such, the disaggregation of striatum-mediated versus GPe-mediated inhibitory fields is achieved based on: lack of rapidly depressing inhibitory evoked field potentials in STN (which receives inhibitory inputs via GPe and not striatum), but a common presence of rapidly depressing evoked field potentials in SNr and GPi (which both receive most of their inhibitory inputs from striatum); differences in the morphology of purportedly GPe- (fast latency) versus striatum-mediated (slow latency) evoked field potentials [16]; and the presence of slow latency caudato-nigral evoked field potentials in slices [19] that are reversed by GABA antagonist application [20]. These points are indeed outlined in the first paragraph of the Discussion sub-section "Direct pathway plasticity." However, we have now additionally added a point to the Limitations that inhibitory inputs to the GPi also come by way of GPe, though in a lesser abundance.

[16] L. A. Steiner et al., "Persistent synaptic inhibition of the subthalamic nucleus by high frequency stimulation," *Brain Stimul*, vol. 15, no. 5, pp. 1223–1232, 2022, doi: 10.1016/j.brs.2022.08.020.

[17] L. D. Liu, I. A. Prescott, J. O. Dostrovsky, M. Hodaie, A. M. Lozano, and W. D. Hutchison, "Frequency-dependent effects of electrical stimulation in the globus pallidus of dystonia patients.," *J Neurophysiol*, vol. 108, no. 1, pp. 5–17, Jul. 2012, doi: 10.1152/jn.00527.2011.

[18] L. Milosevic et al., "Modulation of inhibitory plasticity in basal ganglia output nuclei of patients with Parkinson's disease," *Neurobiology of Disease*, vol. 124, pp. 46–56, Apr. 2019, doi: 10.1016/j.nbd.2018.10.020.

[19] M. Yoshida and W. Precht, "Monosynaptic inhibition of neurons of the substantia nigra by caudato-nigral fibers," *Brain Res*, vol. 32, no. 1, pp. 225–228, Sep. 1971, doi: 10.1016/0006-8993(71)90170-3.

[20] W. Precht and M. Yoshida, "Blockage of caudate-evoked inhibition of neurons in the substantia nigra by picrotoxin," *Brain Res*, vol. 32, no. 1, pp. 229–233, Sep. 1971, doi: 10.1016/0006-8993(71)90171-5.

Amendments to the manuscript:

"Indeed, GPi receives the greatest abundance of inhibitory inputs from striatum (direct pathway), but also it also receives inhibitory inputs by way of GPe (indirect pathway). Although we can functionally disaggregate these pathway-specific responses based on differences in morphology and dynamics of GPe-mediated versus striatum-mediated inhibitory fEPs [21]; the possibility of compounded effects cannot be completely ruled out."

The mechanism of short- and long-term plasticity as applied in the protocols used in this work are outlined in reference to previous citations [15, 16, 18]. Because this is a central aspect of the current work and interpreting the results, it was difficult to appreciate how

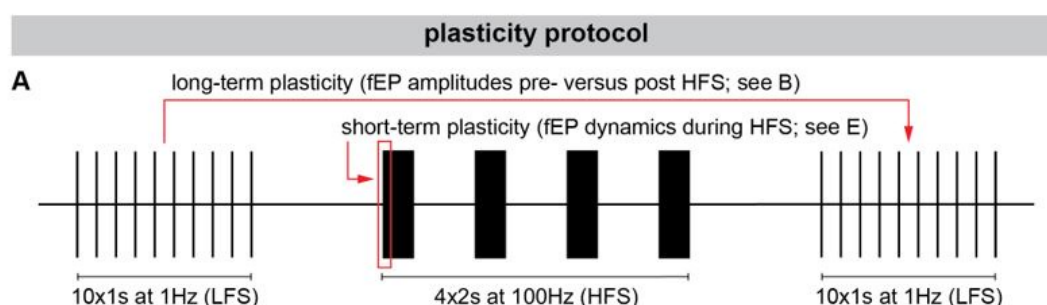
these protocols provide distinct metrics of short and long-term plasticity in GPi without some explanation of how it applies to the current work and the specific mechanisms. It would also help to be able to better link how the results fit with the broader conclusions.

Short-term plasticity is measured as the dynamic change to the fEP during ongoing HFS. For long-term plasticity analyses, the fEP amplitudes during LFS were compared pre- versus post-HFS. To make this analysis more intuitive we have added a protocol illustration to Fig 2. We have moreover greatly expanded the discussion to include more literature related to disease-specific differences in plasticity, and implications of modulating plasticity using DBS.

Amendments to the manuscript:

Added new panel to Fig 2

Author response image 1.



“Converging evidence from past animal and human studies suggests that dystonia is associated with impaired synaptic function and abnormal synaptic plasticity [35–37]. Compared to healthy controls, it has been shown that transcranial magnetic stimulation induced motor evoked potentials (MEPs) are hyperexcitable in dystonia [38,39], and somatosensory and motor cortical plasticity is greater [40]. Likewise, enhanced long-term potentiation at cortico-striatal synapses has been shown in rodent models of dystonia [41,42]. While our finding that long term potentiation effects are greater in PD compared to dystonia (Fig. 2D) is difficult to corroborate with this literature, one potential explanation can be that all of our PD patients are long-term users of levodopa. We have previously shown that the intake of this antiparkinsonian dopaminergic medication leads to potent increases in the amount of plasticity elicited in GPi [15]. Although patients are 12hr withdrawn from antiparkinsonian medications for surgery, it could be that striato-pallidal synapses are nevertheless chronically over-sensitized from prolonged use of dopaminergic medication; which is a well-known hypothesis related to the manifestation of levodopa-induced dyskinesias (a hyperkinetic feature) in PD [43]. Indeed, a lack of depotentiation of striato-pallidal projections has previously been observed in patients with levodopa-induced dyskinesias [44]. As such, excessive plasticity of these projections may corroborate hyperkinetic features of dystonia and levodopa-induced dyskinesias in PD.”

In the Conclusion, it was difficult to understand the sentence about microcircuit interaction (line 232) and how it selectively modulates the efficacy of target synapses. Some further explanation here would be helpful. Also, it was not clear how these investigations (line 237) provide cellular-level support for closed-loop targeting. Could the reference to closed-loop targeting also be further explained?

We agree with the reviewer that the current wording may be confusing. We have changed the wording to be clearer. We have additionally added content related to closed-loop DBS based on chronic monitoring of evoked potential responses.

Amendments to the manuscript:

“Furthermore, chronic monitoring of evoked fields may allow for tracking of subcortical neuronal projections as indexed by inhibitory fields reported in this study. microcircuit interaction to selectively modulate the efficacy of target synapses.”

future applications of DBS may also benefit from closed loop tuning of basal-ganglia-thalamo-cortical circuit dynamics and plasticity through chronic monitoring of evoked potential responses [56].

How is the burst index calculated (Methods)?

Thank you for pointing out that the burst index definition was missing from the paper. It has now been added to the manuscript.

Amendments to the manuscript:

“The burst index was computed by taking the ratio of the means from a two-component Gaussian mixture model applied to the log interspike interval distribution, a modification of the previous mode-over-mean ISI method [20]”

Figures and figure captions are missing some details:

Fig. 1 - What does shading represent?

The shading in Fig. 1 illustrates results that were significant before adjustment for multiple comparisons.

Amendments to the manuscript:

“Depicted scatterplots are results that were significant before correction for multiple comparisons”

Fig. 2 - Can the stimulation artifact be labeled so as not to be confused with the physiological signal? Is A representing the average of all patients or just one example? Are there confidence intervals for this data as it's not clear if the curves are significantly different or not (may not be important to show if just one example)? Same for D. What is being plotted in E? Is this the exponential fitted on data? Can this be stated in the figure citation directly so readers don't have to find it in the text, where it may not be directly obvious which figure the analyses are being applied towards?

Thank you for your comments regarding Fig. 2. We have made the following revisions to address the concerns:

To clarify the presence of stimulation artifacts and differentiate them from the physiological signal, we have updated Panel B and E in the updated Fig. 2 which highlight the stimulation artifacts accordingly.

Regarding the comment about Panel A (now B in the updated figure), it represents one single example per disease, rather than an average of all patients.

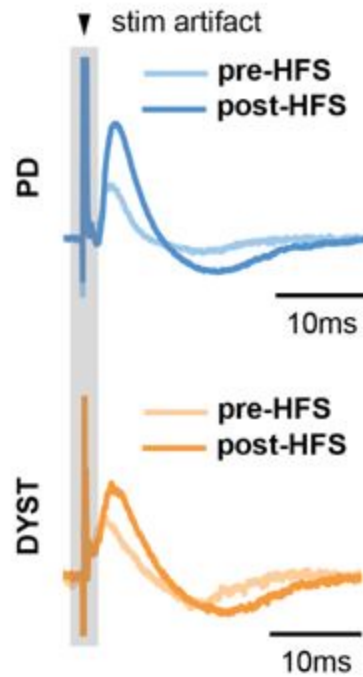
In response to the comment about what is plotted in Panel E, we have revised the figure caption to explicitly state that it includes the exponential fit on the data.

Amendments to the manuscript:

Figure 2 panel B and E now highlight stimulation artifacts.

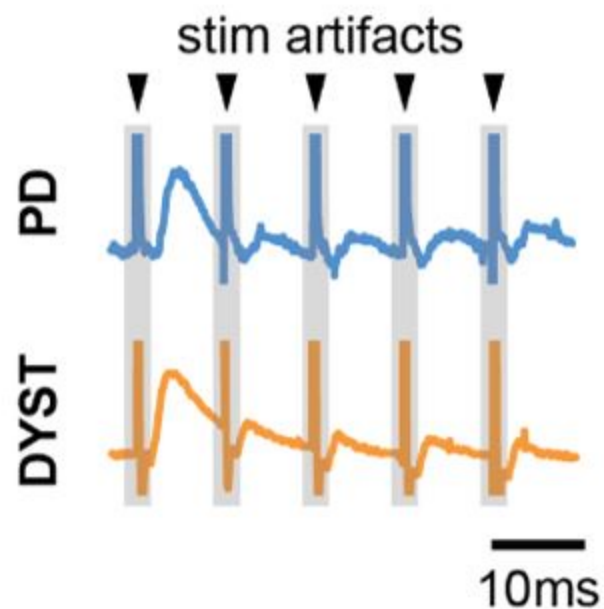
Author response image 2.

B example LFS trace



Author response image 3.

E example HFS trace



The figure captions could use more details, that can be taken from the text, so that readers can understand figures without searching for relevant details across the paper.

Thank you for your feedback. We have revised the figure captions accordingly to provide more details.

Amendments to the manuscript:

“Fig 1 – GPi spiketrain feature analyses and clinical correlates of PD and dystonia. (A) With respect to (A) rate-based spiketrain features, firing rate was greater in PD while burst index (BI) and coefficient of variation (CV) were greater in dystonia; whereas no differences were found for (B) oscillatory spiketrain features for theta, alpha, low beta, high beta frequencies. MWU statistical results depicted are not corrected for multiple comparisons; after correction using the Bonferroni method, only CV and BI results remain significant (please see Supplementary Table 3). (C) In PD, the power of low beta spiketrain oscillations positively correlated (Spearman correlation) with symptom severity; in dystonia, neuronal firing rate negatively correlated with symptom severity, whereas CV and the power of theta spiketrain oscillations positively correlated with symptom severity. Depicted scatterplots are results that were significant before correction for multiple comparisons; however, none of the results persist after Benjamini-Hochberg correction for false discovery rate (please see Supplementary Table 4).”

“Fig 2 – Long-term and short-term effects of HFS on striato-pallidal plasticity in PD and dystonia. (A) Schematic of the plasticity protocol to assess long-term plasticity via fEP amplitude comparisons pre- versus post-HFS and short-term plasticity via fEP dynamics during HFS. (B) Highlights example fEP traces for measuring long-term plasticity pre- versus post-HFS, with (C) displaying group-level fEP amplitudes pre- versus post-HFS across diseases. (D) Illustrates the amount of plasticity (i.e., percentage change in fEP amplitudes pre- versus post-HFS) in both PD and dystonia, with PD showing higher levels of plasticity. (E) Provides an example of fEP traces during HFS for assessing short-term plasticity, with (F) depicting group-level decay rates of fEP amplitudes using an exponential fit on the fEP amplitudes over the first 5 stimulus pulses across diseases. (G) Shows the half-life of the fitted exponential (i.e., rate of attenuation of fEP amplitudes) between PD and dystonia, with PD demonstrating faster fEP attenuation.”

<https://doi.org/10.7554/eLife.90454.2.sa0>